Final Project Report:

Chocolate Regression

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1 Abstract and Introduction

Understanding what affects product sales is important for businesses. It can help them improve their marketing, allow them to manage inventory better, and help them figure out which products sell the most. In this analysis, we will analyze the different factors that affect recorded chocolate sales.

Using a dataset of global chocolate sales transactions from 2022–2024, we study which factors influence the response variable: the amount of money generated from each sale. The goal is to identify the variables that have the most significant impact on sales and to propose an appropriate regression model to explain and predict chocolate revenue.

In this study, we investigate the following questions:

1. What is the best model of chocolate sales, sales, country, year, boxes shipped?
2. Which predictors have the most impact on sales?
3. How well are we able to predict sales using these variables?

2 Data Description and Exploratory Data Analysis (EDA)

This data set was accessed through Kaggle and contains global chocolate sales transactions collected from 2022 to 2024. It consists of $n = 3013$ observations and 5 columns/variables. Each observation represents a single chocolate sales transaction. The data set includes both quantitative and categorical variables related to each sale.

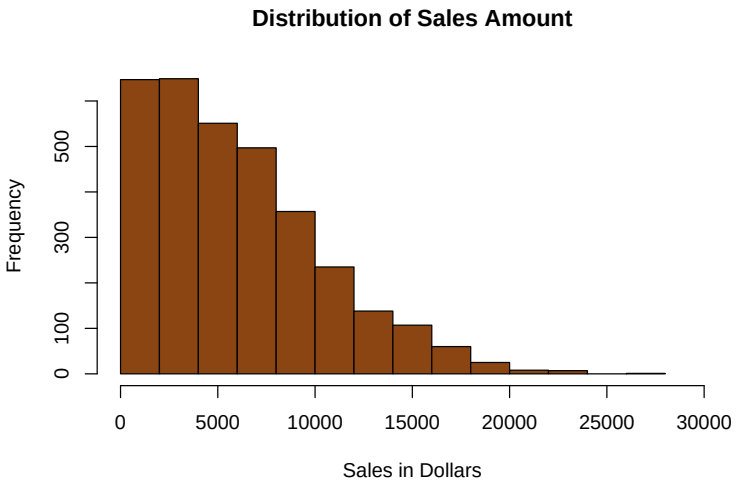
2.1 Variables

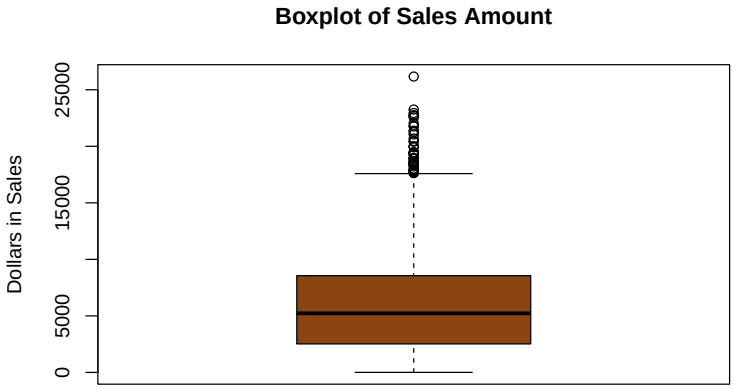
Let Y denote the response variable representing chocolate sales, and let X_1–X_4 denote the predictors:

- Y: sqrt_Amount – square root transformation of sales amount (USD) for each transaction.
- X_1: Product – type of chocolate product sold (e.g., dark chocolate bites, caramel chocolates, eclairs).
- X_2: Country – country where the sale occurred.
- X_3: Year – year of the transaction, derived from the Date variable.
- X_4: Boxes.Shipped – number of chocolate boxes sold in the transaction.

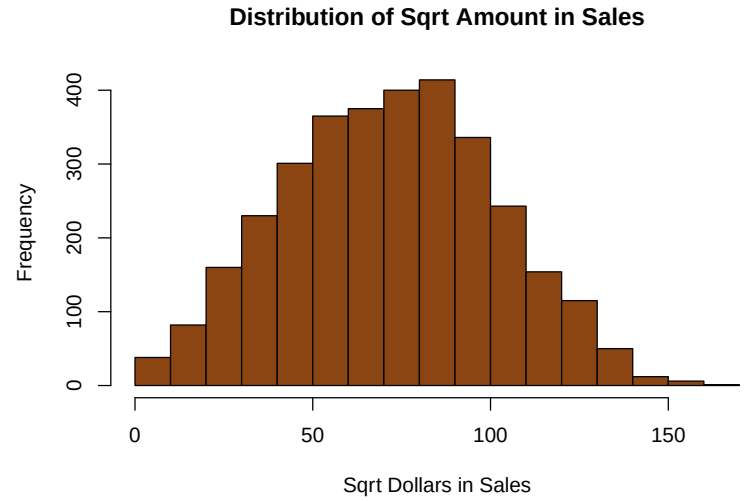
2.2 Data Cleaning

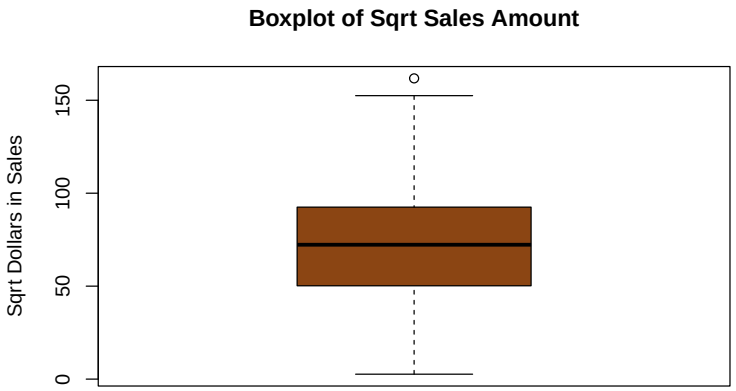
Our first step was to make the `Amount` and `Date` columns numerical. Next, we plotted the distribution of `Amount`:





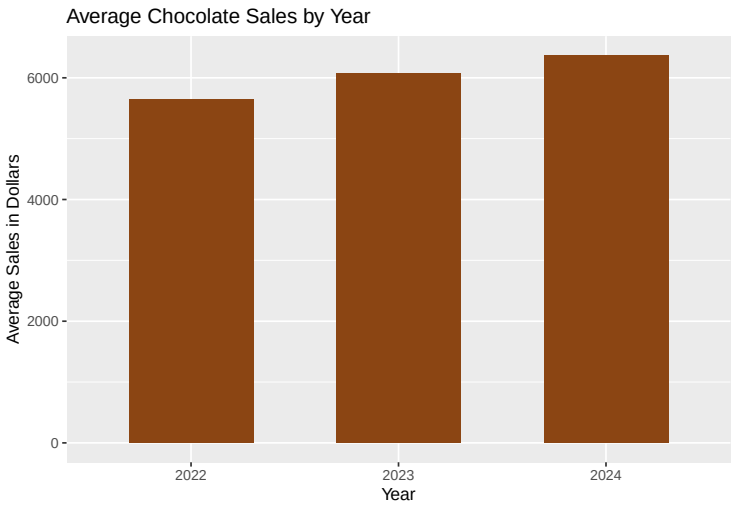
The Distribution of `Amount` was originally right skewed. The skew of the distribution of the response variable can impact the distribution of the residuals, which need to be normally distributed to satisfy model assumptions. To transform it, we took the square root of Amount, which yielded a normal distribution:

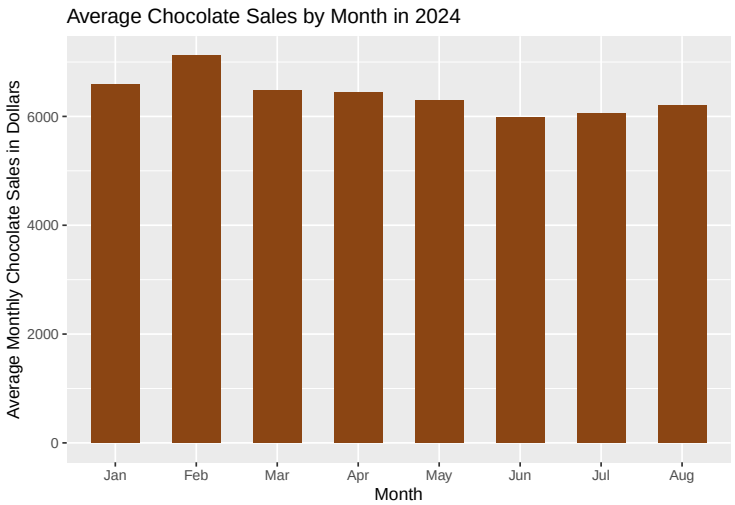




2.3 Data Exploration

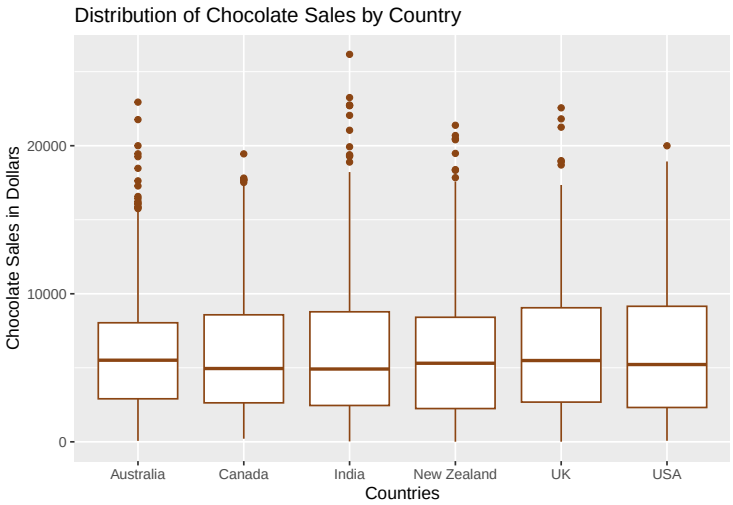
a) Does average sales Amount differ by year?

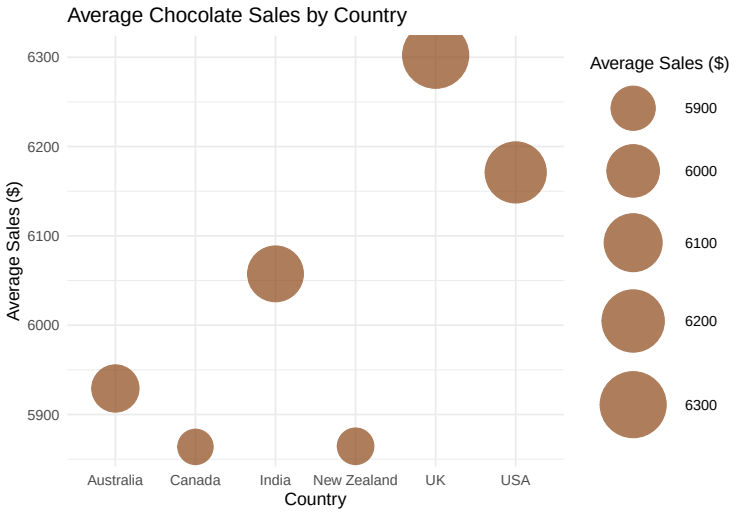




Sales rose from about $5652.308 in 2022 to roughly $6366.151 in 2024. This suggests consistent growth in chocolate sales over the three-year period. Regarding average monthly sales, February has the highest average sales, likely reflecting seasonal demand such as Valentine's Day (love is in the air)!

b) How do chocolate sales vary by country?

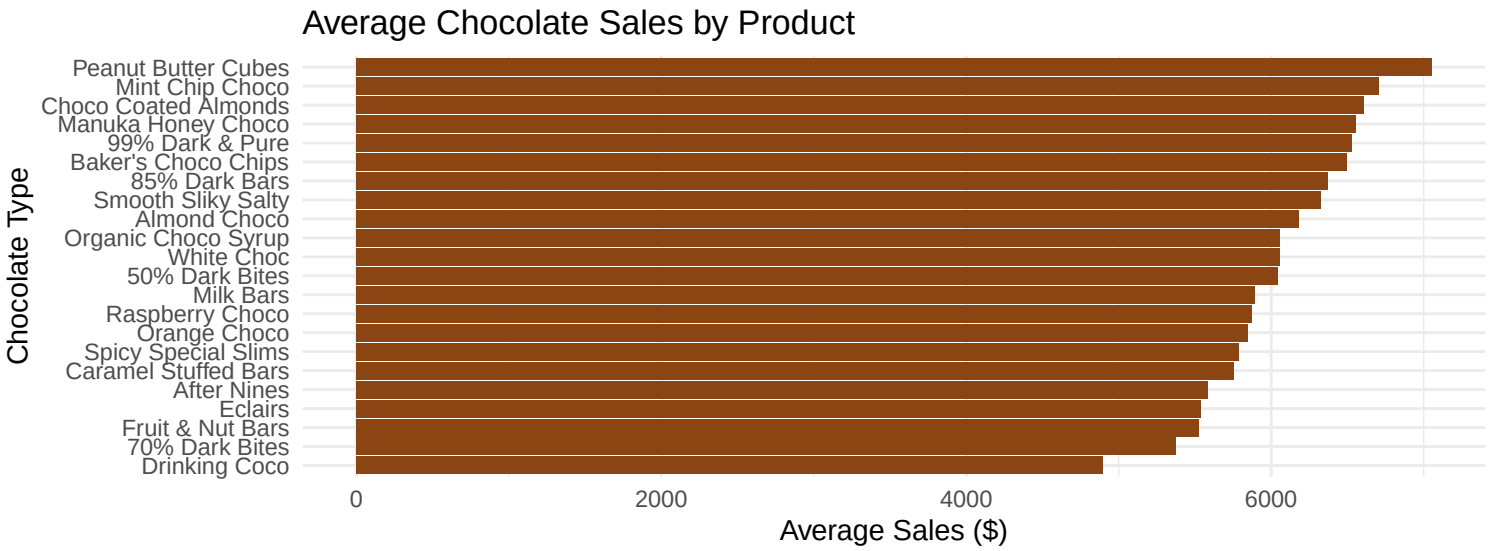




The UK, USA, and Australia tend to have slightly higher median sales compared to other countries. India also has higher outliers.

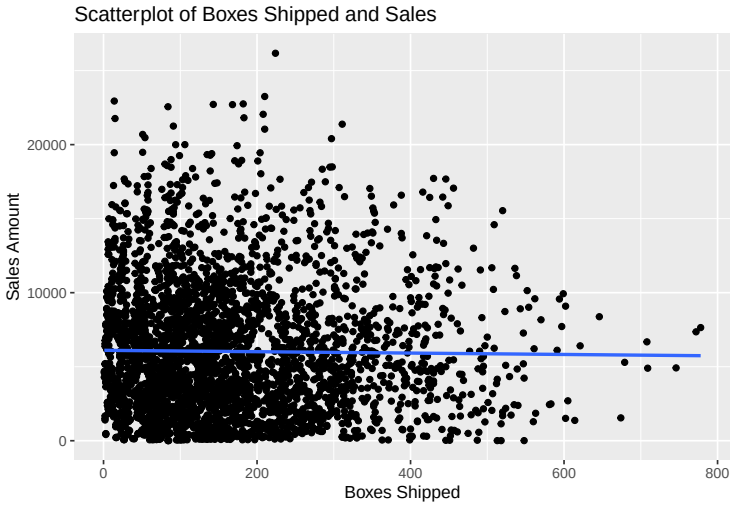
The UK stands out with the highest average sales, while Canada and New Zealand have lower averages.

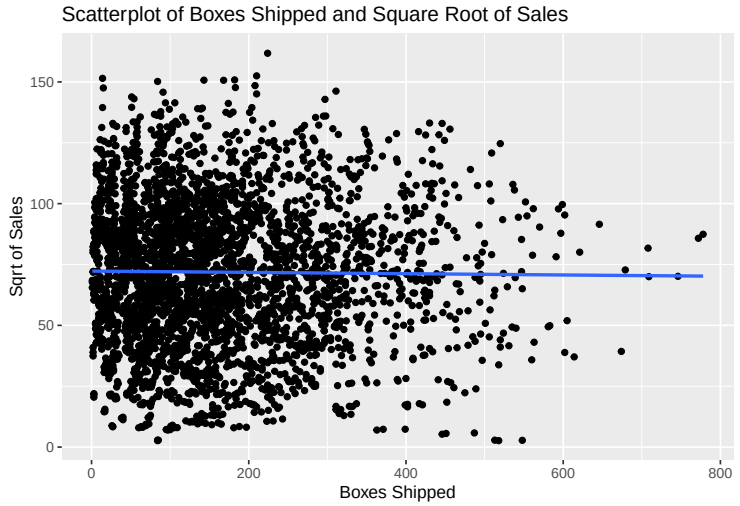
c) How do chocolate sales vary by product?



Peanut Butter Cubes have the highest average sales, while Drinking Coco has the lowest.

d) Why is there no correlation between `Boxes.Shipped` and `Amount`?





There is no clear correlation between `Boxes.Shipped` and `Amount` for each order (with or without transformation) even though theoretically, purchasing more boxes should increase the total sales amount for a transaction.

This likely reflects differences in product prices. Each transaction records only one type of chocolate product, and orders containing

multiple products appear as separate transactions in the dataset, even if they occur on the same date. To better understand price differences across products and countries, we created a new variable, *price_per_box*:

Table 1: Average price per box by country

Country	observations	avg_price_per_box
Canada	525	127.34474
USA	537	125.44095
New Zealand	519	124.61316
Australia	615	100.62601
India	552	97.62574
UK	534	95.04117

Table 2: Products with the highest and lowest average price per box

Product	observations	avg_price_box
White Choc	174	246.48764
70% Dark Bites	126	61.32908

The most expensive chocolate product is a box of white chocolate at about $246.49, with the least expensive being the 70% dark bites at a price of about $61.33. If one transaction represented a purchase of 15 boxes of white chocolate, and another transaction represented a purchase of 30 boxes of 70% dark bites, the first purchase would still have a larger sales amount. This may indicate the reasoning for the lack of correlation between `Boxes.Shipped` and `Amount`.

3 Methodology and Model Building

3.1 Single Predictor Models

For each single predictor model, we recorded the p-value, R^2 adjusted, and whether or not the independent predictor was significant.

Table 3: P-values for Single Predictor Models

Predictor	P-value	Significant?	R^2 adj
Boxes.Shipped	0.5246	No	-0.0001814
Country	0.6509	No	-0.0005125
Product	0.0033	Yes	0.0066634
Year	0.0031	Yes	0.0029044
price_per_box	<2e-16	Yes	0.0552758

Although we included `price_per_box` in this table, we actually cannot include it in our model since price_per_box $\times$ Boxes.Shipped = Amount. This will create complications and multicollinearity in our model, so we must omit `price_per_box`.

The p-value results from the single predictor models indicate that several predictors have a significant impact on the response variable, specifically `Product` and `Year`. Although `Boxes.Shipped` and `Country` are individually insignificant, they should still be evaluated in multivariable models, as they may become significant when combined with other predictors.

3.2 Stepwise Selection

Stepwise selective works by adding or removing predictor variables to achieve the best fit model. In our case, backward, forward, and both-direction methods resulted in the same pick.

Table 4: Comparison of Stepwise Selection Procedures

Stepwise Method	Final Predictors	AIC Before Stepwise	AIC After Stepwise
Forward	Product + Year	22207.77	21999.69
Backward	Product + Year	22008.42	21999.69
Both	Product + Year	22008.42	21999.69

Based off these results we know that best model according to Stepwise Selection is:

$$\sqrt{Amount} \sim \text{Product} + \text{Year}$$

3.3 Adding Interaction Terms

Full model with no Interaction Variables

$$\sqrt{Amount} \sim \text{Boxes.Shipped} + \text{Product} + \text{Year} + \text{Country}$$

Considering Interaction Variables:

To determine which interaction terms were reasonable, we interacted the stepwise selected predictors with all other predictors to find where there were significant interactions.

The p-value in the table below is the resulting p-value from performing an ANOVA test with the stepwise model **including** that interaction term and the original stepwise model.

Table 5: P-values for Interaction Term Models

Interaction Term	ANOVA p-value	Significant?
Full Model No Interactions	0.726	No
Product:Boxes.Shipped	7.672e-07	Yes
Product:Country	$<$ 2.2e-16	Yes
Product:Year	0.0025	Yes
Year:Boxes.Shipped	0.9127	No
Year:Country	0.9995	No
Country:Boxes.Shipped	0.01642	Yes

The ANOVA test p-value indicates that there is a significant impact from several interaction terms, specifically `Product:Boxes.Shipped`, `Product:Country`, `Product:Year`, and `Country:Boxes.Shipped`.

3.4 Proposed Model

Based on which interaction terms were significant, we propose a tentative final model of the stepwise pick with all the significant interaction terms:

$$\sqrt{\text{Amount}} \sim \text{Product} + \text{Year} + \text{Product:Boxes.Shipped} + \text{Product:Country} + \text{Product:Year} + \text{Country:Boxes.Shipped}$$

We are not sure if including `Country:Boxes.Shipped` is a good choice since it was less significant than other terms. To ensure we are not making a mistake, we will test our proposed final model AIC, BIC, and Adjusted R^2 with `Country:Boxes.Shipped` and without:

Table 6: Comparison of AIC and BIC With/Without Country:Boxes.Shipped

Model	Adjusted R2	AIC	BIC
Model Without Country:Boxes.Shipped	0.0681	31483.11	32696.26
Final Model	0.0717	31474.78	32718.41

The difference between the two models is not very big, but since the final model maximizes adjusted R^2 and minimizes AIC, we will choose this model. The BIC is minimized for the model without the extra interaction, but we also know that BIC tends to be bias towards simpler models.

Our proposed model is now our final model!

3.5 Model Summary

Here are the most significant (p-value <= 0.01) coefficients:

Table 7: Estimated Regression Coefficients and Significance

	Predictor	Estimate	Std. Error	t value	Pr(>|t|)
1	(Intercept)	68.2160	6.0450	11.2848	0.0000
33	ProductChoco Coated Almonds:Boxes.Shipped	-0.1029	0.0292	-3.5207	0.0004
48	Product70% Dark Bites:CountryCanada	24.1449	8.5902	2.8108	0.0050
57	ProductEclairs:CountryCanada	-35.0116	8.1791	-4.2806	0.0000
69	Product50% Dark Bites:CountryIndia	20.1749	7.5426	2.6748	0.0075
84	ProductOrange Choco:CountryIndia	-25.5584	8.9800	-2.8462	0.0045
85	ProductOrganic Choco Syrup:CountryIndia	23.6312	7.7276	3.0580	0.0022
111	ProductSpicy Special Slims:CountryNew Zealand	-24.7824	9.1399	-2.7115	0.0067
119	ProductBaker's Choco Chips:CountryUK	-24.7060	8.5349	-2.8947	0.0038
150	ProductOrange Choco:CountryUSA	-22.2072	8.2116	-2.7044	0.0069
155	ProductSpicy Special Slims:CountryUSA	-41.9466	8.9889	-4.6665	0.0000

Summarize Important Coefficients:

In total, the final model has 203 coefficients due to having 22 products that are considered in 3 interaction terms. We found that we had four significant coefficients from interacting Product with Boxes.Shipped, 19 significant coefficients from interacting Product with Country, and one significant coefficient from interacting Boxes.Shipped with Country.

We'll interpret one of these coefficients from each category due to time and space limitations.

Interpretation of the Intercept: The square root of Amount for the base line product, 50% Dark Bites, shipped to the reference country, Australia, in the base line year, 2022, assuming that all variables are held constant is 68.22 dollars.

Boxes.Shipped:CountryNew Zealand Holding all other variables constant, for every one unit increase in Boxes.shipped, the square root of Amount decreases 0.0324 dollars for shipments to New Zealand than for the reference country which is Australia.

ProductRaspberry Choco:Boxes.Shipped Holding all other variables constant, for the flavor Raspberry Chocolate, each one unit increase in Boxes.Shipped is associated with a 0.0611 unit increase in the square root of Amount compared to the the reference product of 50% Dark Bites.

ProductRaspberryChoco:CountryUSA Holding all other variables constant, the square root of Amount for Raspberry Choco in the USA is 14.21 dollars higher than expected. This is comparing the expected values of Raspberry Choco and USA's effects seperately in a model and relative to the baseline combination which is 50% Dark Bites in Australia.

ProductRaspberryChoco:Year2023 Holding all other variables constant, the square root of Amount for Raspberry Choco in the year 2023 is 0.0689 units lower than expected when the effects of Product and Year are separate for Raspberry Choco and year 2023. This would be relative to the baseline year 2022, and the baseline product 50% Dark Bites.

4 Results and Diagnostics

4.1 AIC BIC Comparison

Compare AIC and BIC for Stepwise Selected model and final model:

Table 8: Comparison of AIC/BIC for Stepwise and Final Model

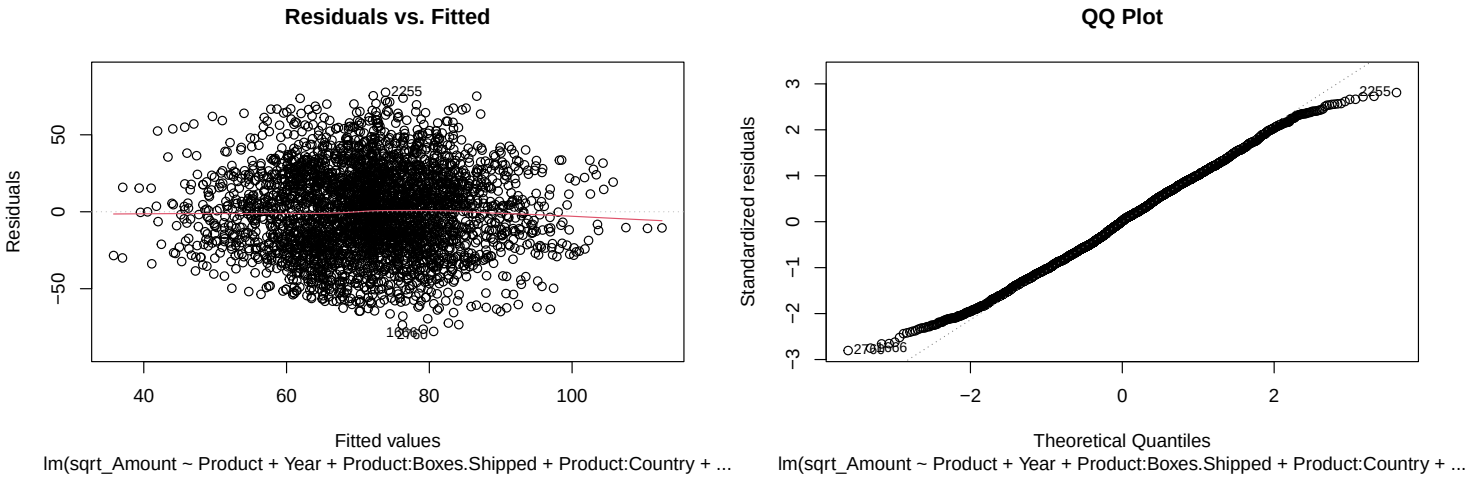
Model	AIC	BIC
Stepwise Model	31514.97	31667.37
Final Model	31474.78	32718.41

AIC is minimized for our final model. This is a good sign for the quality of this model. BIC is minimized for the simpler model, which is expected since BIC penalizes model complexity (number of predictors) more.

4.2 Model Diagnostics

Let's check that this model satisfies the assumptions:

$$\epsilon_i \sim^{iid} Norm(0, \sigma^2)$$



Residuals vs Fitted plot shows that homoscedasticity is a valid assumption. Variance is regular throughout the residuals.

Because our QQ-plot shows a linear line, residuals are also normally distributed in this model.

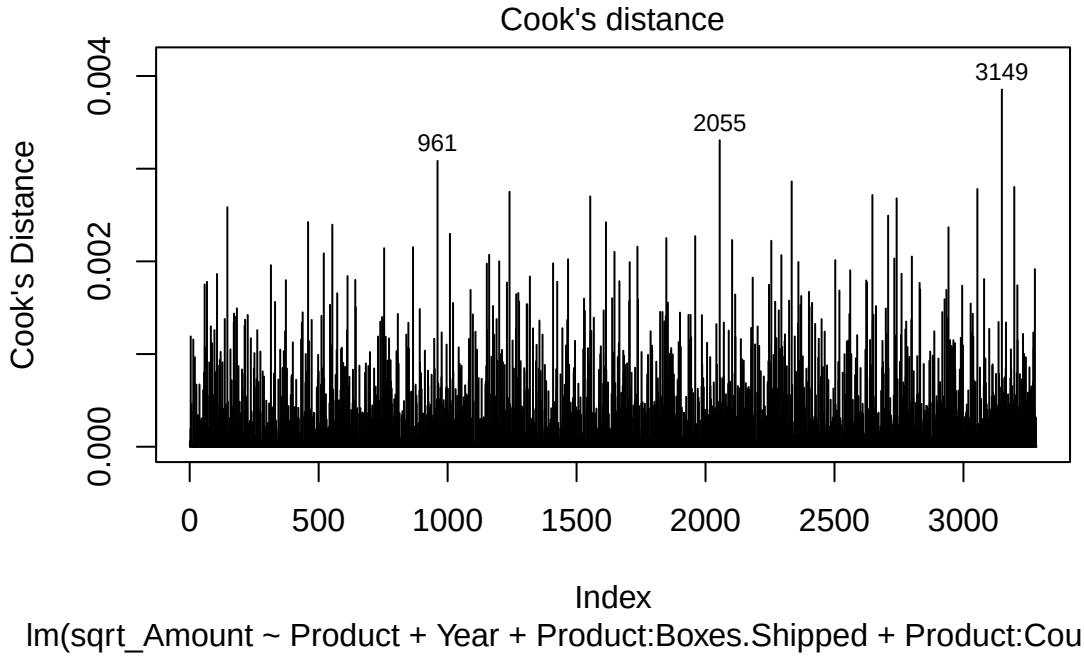
Check for Multicollinearity

Table 9: Variance Inflation Factors for Predictors in the Final Model

Predictor	Other Predictors Included	VIF
Product	Product:Boxes.Shipped, Product:Country, Product:Year	1.0000
Year	Product:Year	1.3887
Boxes.Shipped	Product:Boxes.Shipped, Country:Boxes.Shipped	1.0758
Country	Product:Country, Country:Boxes.Shipped	1.0758

Under the $GVIF^{(1/(2*Df))}$, we can tell that all VIF's are approximately 1, and even the largest value (1.39 for Year) is below concerning thresholds. The predictors in the model do not exhibit problematic multicollinearity.

Check for Outliers



Most observations have very small Cook's Distance values close to 0, with the highest at 0.004, which means removing them would not meaningfully change the regression model as they do not exert high leverage.

5 Discussion and Conclusion

Considering the significance of predictors and interactions, AIC, and Adjusted R^2:

$$\sqrt{\text{Amount}} \sim \text{Product} + \text{Year} + \text{Product:Boxes.Shipped} + \text{Product:Country} + \text{ Product:Year} + \text{ Country:Boxes.Shipped}$$

Based on our tests, `Product` and `Year` had the strongest individual effects on sales. However, `Country` and `Boxes.Shipped` became significant when interacting with other variables, particularly `Product`. This suggests that the effect of shipment size and location on sales may vary depending on the product being sold. Despite these interactions, the VIF values were low, indicating that multicollinearity is not a concern in the model.

One of our challenges involved the *price_per_box* variable, which initially seemed highly significant. Once we realized that it was calculated directly from the response variable (`Amount/Boxes.Shipped`), including it in the regression would make it illogical and misrepresent the relationship. Thus, *price_per_box* was excluded from the final model.

There were several limitations in the dataset. First, the model included only one numerical predictor (`Boxes.Shipped`), and this variable was not strongly correlated with the response on its own. Additionally, the dataset only covers transactions from 2022–2024 and includes months from January through August, limiting long-term or full year patterns. Another limitation of the model is its low explanatory power (even the final model has an adjusted R^2 of only about 0.0717!), suggesting that other variables not included in the dataset likely influence sales.

One potential improvement would be to fit separate models for each country. Because product demand and pricing may differ across countries, country-specific models could better capture these relationships. For example, we fit a model using only data from Australia (country with largest number of observations) resulted in a much lower AIC and a slightly higher R^2.

Model	R_squared	AIC
Australia Only Model	0.1034615	5861.276
All Countries Model	0.0717491	31474.779

Appendix

```
chocolate <- read.csv("chocolate.csv")
head(chocolate)
```

```
#Transform "Amount" (using gsub)
chocolate$Amount <- as.numeric(gsub("[$,]", "", chocolate$Amount))
#Making Year category
chocolate$Year <- substring(chocolate$Date,7,10)
head(chocolate)
```

```
#Histogram of Sales

hist(chocolate$Amount, col = "chocolate4",
     main = "Distribution of Sales Amount",
     xlab = "Sales in Dollars", xlim = c(0,30000))
#Box plot of Sales

boxplot(chocolate$Amount,
        main = "Boxplot of Sales Amount",
        ylab = "Dollars in Sales",
        col = "chocolate4")
```

```
#Square Root Transformation of Sales

chocolate$sqrt_Amount <- sqrt(chocolate$Amount)

#Histogram of Sqrt Sales

hist(chocolate$sqrt_Amount, col = "chocolate4",
     main = "Distribution of Sqrt Amount in Sales",
     xlab = "Sqrt Dollars in Sales")

#Box plot of sqrt of Sales

boxplot(chocolate$sqrt_Amount,
        main = "Boxplot of Sqrt Sales Amount",
        ylab = "Sqrt Dollars in Sales",
        col = "chocolate4")
```

```
#Bar Plot of Average Chocolate sales in 2022, 2023, and 2024

average_sales <- chocolate %>%
  group_by(Year) %>%
  summarise(average_sales = mean(Amount, na.rm = TRUE))

#average_sales


ggplot(average_sales, aes(x = factor(Year), y = average_sales)) +
  geom_col(fill = "chocolate4", width = 0.6) +
  labs(title = "Average Chocolate Sales by Year",
       x = "Year",
       y = "Average Sales in Dollars")



# Chocolate Sales Monthly

chocolate_2024 <- chocolate %>%
  filter(Year == 2024)
```

```r
#Making Month Category in 2024
chocolate_2024$Month <- substring(chocolate_2024$Date, 4, 5)

chocolate_2024$Month <- factor(chocolate_2024$Month,
                               levels = c("01","02","03","04","05","06","07","08","09","10","11","12"),
                               labels = c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"

# Average Sales of Chocolate by Month in 2024

average_sales_2024 <- chocolate_2024 %>%
  group_by(Month) %>%
  summarise(average_sales_2024 = mean(Amount, na.rm = TRUE))
#average_sales_2024

#Bar Plot of Monthly Chocolate Sales in 2024

ggplot(average_sales_2024, aes(x = Month, y = average_sales_2024)) +
  geom_col(fill = "chocolate4", width = 0.6) +
  labs(title = "Average Chocolate Sales by Month in 2024",
       x = "Month",
       y = "Average Monthly Chocolate Sales in Dollars")
```

```r
# Average Country Sales
average_country_sales <- chocolate %>%
  group_by(Country) %>%
  summarise(average_country_sales = mean(Amount, na.rm = TRUE)) %>%
  arrange(desc(average_country_sales))


#average_country_sales

# Box Plot for Country vs Chocolate Sales

ggplot(chocolate, aes(x = Country, y = Amount)) +
  geom_boxplot(color = "chocolate4") +
  labs(title = "Distribution of Chocolate Sales by Country",
       x = "Countries",
       y = "Chocolate Sales in Dollars")



ggplot(average_country_sales, aes(x = Country, y = average_country_sales, size = average_country_sales)) +
  scale_size(range = c(10, 19)) +
  geom_point(color = "chocolate4", alpha = 0.7) +
  scale_color_gradient(low = "chocolate4", high = "chocolate4") +
  labs(title = "Average Chocolate Sales by Country",
       x = "Country",
       y = "Average Sales ($)",
       size = "Average Sales ($)") +
  theme_minimal()
```

```r
num <- length(unique(chocolate$Product))

# Average Product Sales
average_product <- chocolate %>%
  group_by(Product) %>%
  summarise(average_product = mean(Amount, na.rm = TRUE)) %>%
  arrange(desc(average_product))

#average_product
```

```r
ggplot(average_product, aes(x = reorder(Product, average_product), y = average_product)) +
  geom_col(fill = "chocolate4") +
  coord_flip() +
  labs(title = "Average Chocolate Sales by Product",
       x = "Chocolate Type",
       y = "Average Sales ($)") +
  theme_minimal()
```

```r
#Scatterplot of Boxes Shipped and Sales
ggplot(data = chocolate, aes(x = Boxes.Shipped, y = Amount)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Scatterplot of Boxes Shipped and Sales",
       x = "Boxes Shipped",
       y = "Sales Amount")


#Sqrt of Sales and Boxes Shipped
ggplot(data = chocolate, aes(x = Boxes.Shipped, y = sqrt_Amount)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Scatterplot of Boxes Shipped and Square Root of Sales",
       x = "Boxes Shipped",
       y = "Sqrt of Sales")
```

```r
chocolate$price_per_box <- chocolate$Amount / chocolate$Boxes.Shipped

# Table of countries and observations
countries_obs <- chocolate %>%
  group_by(Country) %>%
  summarise(observations = n(),
            avg_price_per_box = mean(price_per_box, na.rm = TRUE)) %>%
  arrange(desc(avg_price_per_box))

knitr::kable(
  countries_obs,
  caption = "Average price per box by country"
)
```

```r
# Table of product and price_per_box
product_prices <- chocolate %>%
  group_by(Product) %>%
  summarise(observations = n(),
            avg_price_box = mean(price_per_box, na.rm = TRUE)) %>%
  arrange(desc(avg_price_box))

extremes <- product_prices %>%
  slice(c(1, n()))

knitr::kable(
  extremes,
  caption = "Products with the highest and lowest average price per box"
)
```

```r
model_boxes <- lm(sqrt_Amount ~ Boxes.Shipped, data = chocolate)

summary(model_boxes)
```

```r
model_country <- lm(sqrt_Amount ~ Country, data = chocolate)

summary(model_country)
```

```r
model_product <- lm(sqrt_Amount ~ Product, data = chocolate)

summary(model_product)
```

```r
model_year <- lm(sqrt_Amount ~ Year, data = chocolate)

summary(model_year)
```

```r
model_price_per <- lm(sqrt_Amount ~ price_per_box, data = chocolate)
summary(model_price_per)
```

```r
# Extract p-values
p_boxes   <- summary(model_boxes)$coefficients["Boxes.Shipped","Pr(>|t|)"]
p_country <- anova(model_country)$`Pr(>F)`[1]
p_product <- anova(model_product)$`Pr(>F)`[1]
p_year    <- anova(model_year)$`Pr(>F)`[1]
p_price   <- summary(model_price_per)$coefficients["price_per_box","Pr(>|t|)"]

r_boxes <- summary(model_boxes)$adj.r.squared
r_country <- summary(model_country)$adj.r.squared
r_product <- summary(model_product)$adj.r.squared
r_year <- summary(model_year)$adj.r.squared
r_price <- summary(model_price_per)$adj.r.squared

# Create table
pvalue_table <- data.frame(
  Predictor = c("Boxes.Shipped", "Country", "Product", "Year", "price_per_box"),
  P_value = c(
    round(p_boxes,4),
    round(p_country,4),
    round(p_product,4),
    round(p_year,4),
    "<2e-16"
  ),
  Significant = c("No", "No", "Yes", "Yes", "Yes"),
  Adj_R = c(
    r_boxes,
    r_country,
    r_product,
    r_year,
    r_price
  )
)

knitr::kable(pvalue_table,
             col.names = c("Predictor", "P-value", "Significant?", "R^2 adj"),
             caption = ("P-values for Single Predictor Models"))
```

```r
#Backwards
full_model <- lm(sqrt_Amount ~ Boxes.Shipped + Product + Year + Country,
                 data = chocolate)
backward_model <- step(full_model, dir = "backward")
print(backward_model)
```

```r
#Forwards
empty_model <- lm(sqrt_Amount ~ 1, data = chocolate)

# empty model
forward <- step(empty_model, dir = "forward", scope = list(upper=full_model, lower=empty_model))
print(forward)
```

```r
#both
both <- step(full_model, dir = "both")
print(both)
```

```r
stepwise_summary <- data.frame(
  Stepwise_Method = c("Forward", "Backward", "Both"),
  Final_Predictors = c(
    "Product + Year",
    "Product + Year",
    "Product + Year"
  ),
  AIC_Before = c(22207.77, 22008.42, 22008.42),
  AIC_After = c(21999.69, 21999.69, 21999.69)
)

knitr::kable(
  stepwise_summary,
  col.names = c("Stepwise Method",
                "Final Predictors",
                "AIC Before Stepwise",
                "AIC After Stepwise"),
  caption = "Comparison of Stepwise Selection Procedures"
)
```

```r
step_model = lm(sqrt_Amount ~ Product + Year, data = chocolate)
```

```r
full <- lm(sqrt_Amount ~ Boxes.Shipped + Product + Year + Country, data = chocolate)

anova(full_model, step_model)
```

```r
int1_step_model <- lm(sqrt_Amount ~ Product + Year + Product:Boxes.Shipped,
                      data = chocolate)

anova(step_model, int1_step_model)
```

```r
int2_step_model <- lm(sqrt_Amount ~ Product + Year + Product:Country,
                      data = chocolate)

anova(step_model, int2_step_model)
```

```r
int3_step_model <- lm(sqrt_Amount ~ Product + Year + Product:Year,
                      data = chocolate)

anova(step_model, int3_step_model)
```

```r
int4_step_model <- lm(sqrt_Amount ~ Product + Year + Year:Boxes.Shipped, data = chocolate)

anova(step_model, int4_step_model)
```

```r
int5_step_model <- lm(sqrt_Amount ~ Product + Year + Year:Country,
                      data = chocolate)

anova(step_model, int5_step_model)
```

```r
int6_step_model <- lm(sqrt_Amount ~ Product + Year + Country:Boxes.Shipped,
                      data = chocolate)


anova(step_model, int6_step_model)
```

```r
interaction_table <- data.frame(
  Interaction = c(
    "Full Model No Interactions",
```

```
    "Product:Boxes.Shipped",
    "Product:Country",
    "Product:Year",
    "Year:Boxes.Shipped",
    "Year:Country",
    "Country:Boxes.Shipped"
  ),
  P_value = c(
    "0.726",
    "7.672e-07",
    "< 2.2e-16",
    "0.0025",
    "0.9127",
    "0.9995",
    "0.01642"
  ),
  Significant = c(
    "No",
    "Yes",
    "Yes",
    "Yes",
    "No",
    "No",
    "Yes"
  )
)

knitr::kable(
  interaction_table,
  col.names = c("Interaction Term", "ANOVA p-value", "Significant?"),
  caption = "P-values for Interaction Term Models"
)
```

```
potential <- lm(sqrt_Amount ~ Product + Year + Product:Boxes.Shipped
                + Product:Country + Product:Year + Country:Boxes.Shipped,
                data = chocolate)
```

```
potential_noboxcountry <- lm(sqrt_Amount ~ Product + Year + Product:Boxes.Shipped
                             + Product:Country + Product:Year,
                             data = chocolate)

maybe_final <- data.frame(
  Model = c("Model Without Country:Boxes.Shipped", "Final Model"),
  Adjusted_R2 = round(c(summary(potential_noboxcountry)$adj.r.squared,    summary(potential)$adj.r.squared),4),
  AIC = c(AIC(potential_noboxcountry), AIC(potential)),
  BIC = c(BIC(potential_noboxcountry), BIC(potential))
)

knitr::kable(
  maybe_final,
  col.names = c("Model","Adjusted R2", "AIC", "BIC"),
  caption = "Comparison of AIC and BIC With/Without Country:Boxes.Shipped"
)
```

```
FINAL_model <- lm(sqrt_Amount ~ Product + Year + Product:Boxes.Shipped +
                  Product:Country + Product:Year + Country:Boxes.Shipped,
                  data = chocolate)
```

```
coef_table <- as.data.frame(summary(FINAL_model)$coefficients)

coef_table$Predictor <- rownames(coef_table)
```

```r
coef_table <- coef_table[, c("Predictor","Estimate","Std. Error","t value","Pr(>|t|)")]

rownames(coef_table) <- NULL

# round all numbers to 4 decimals
coef_table[, sapply(coef_table, is.numeric)] <-
  round(coef_table[, sapply(coef_table, is.numeric)], 4)

# get significant predictors only
coef_table <- coef_table[coef_table$`Pr(>|t|)` <= 0.01, ]

knitr::kable(coef_table,
             caption = "Estimated Regression Coefficients and Significance")
```

```r
model_stats <- data.frame(
  Model = c("Stepwise Model", "Final Model"),
  AIC = c(AIC(step_model), AIC(FINAL_model)),
  BIC = c(BIC(step_model), BIC(FINAL_model))
)

knitr::kable(
  model_stats,
  col.names = c("Model", "AIC", "BIC"),
  caption = "Comparison of AIC/BIC for Stepwise and Final Model"
)
```

```r
plot(FINAL_model, which=1, caption="", main="Residuals vs. Fitted")
plot(FINAL_model, which=2, caption="", main="QQ Plot")
```

```r
vif_vals <- vif(FINAL_model, type = "predictor")

vif_table <- data.frame(
  Predictor = rownames(vif_vals),
  Terms_Included = c(
    "Product:Boxes.Shipped, Product:Country, Product:Year",
    "Product:Year",
    "Product:Boxes.Shipped, Country:Boxes.Shipped",
    "Product:Country, Country:Boxes.Shipped"
  ),
  VIF = round(vif_vals[, "GVIF^(1/(2*Df))"], 4)
)

knitr::kable(
  vif_table,
  col.names = c("Predictor", "Other Predictors Included", "VIF"),
  caption = "Variance Inflation Factors for Predictors in the Final Model"
)
```

```r
plot(FINAL_model, which = 4, ann= FALSE)
title(ylab = "Cook's Distance", main = "", xlab = "Index")
```

```r
choc_aus <- chocolate %>%
  filter(Country == "Australia")

modelaus <- lm(sqrt_Amount ~ Product + Year + Product:Boxes.Shipped
               + Product:Year, data = choc_aus)

model_aus_stats <- data.frame(
  Model = c("Australia Only Model", "All Countries Model"),
  R_squared = c(summary(modelaus)$adj.r.squared, summary(FINAL_model)$adj.r.squared),
  AIC = c(AIC(modelaus), AIC(FINAL_model))
)
```

```r
knitr::kable(
  model_aus_stats
)
```